SAM-GC: Robust Automatic Co-segmentation via Mask-Aware Graph Consensus

EE655: Computer Vision and Deep Learning – Course Project
Instructor: Dr. Koteswar Rao Jerripothula

Prince Yadav	Pravir Mishra	Khushi Jain	Vidhi Vivek Chordia	Kanishk Dhariwal
Roll No. 230792	Roll No. 230787	Roll No. 230560	Roll No. 231147	Roll No. 230111
princey23@iitk.ac.in	[Email]	[Email]	[Email]	[Email]

Abstract

*While the Segment Anything Model (SAM) has redefined promptable segmentation, its application in automatic co-segmentation is hindered by a lack of semantic guidance. We present **SAM-GC**, a framework that automates the discovery of common objects across unconstrained image sets. Our primary contribution is a **Mask-Aware Consensus Solver** that reformulates co-segmentation as a graph-based energy minimization problem. By integrating DINOv2 semantic embeddings with geometric priors—specifically log-ratio area compatibility and boundary-fragment penalties—our solver effectively filters category-agnostic noise from SAM proposals. Experimental results demonstrate that SAM-GC achieves high group cohesion and robust zero-shot labeling via CLIP, providing a scalable solution for unsupervised object discovery in unconstrained environments.*

1. Introduction

The task of image co-segmentation—identifying and segmenting the same semantic object across a collection of disparate images—remains a significant challenge in unsupervised computer vision. While the Segment Anything Model (SAM) has revolutionized the field with its high-fidelity, class-agnostic masks, its standard "automatic" mode lacks the inter-image communication necessary to identify a common entity across a group.

Our project, **SAM-GC**, addresses this gap by introducing a graph-based consensus layer that sits atop the SAM architecture. By leveraging self-supervised features from DINOv2 and semantic grounding from CLIP, we move from simple "segment everything" to "segment the common object."

1.1. Motivation and Novelty

In many real-world scenarios, such as product cataloging or wildlife monitoring, we possess groups of images containing a shared subject but lack the manual prompts (points or boxes) required to trigger SAM. Our proposed method automates this by treating every generated mask as a candidate node in a global graph. The novelty of our approach lies in the **weighted consensus solver** located in `graph_solver.py`, which filters out background noise using geometric priors and semantic affinity.

1.2. Team Contributions

This project was a collaborative effort. The specific contributions of our team members are detailed below:

- **Prince Yadav (230792):** [Placeholder for individual contribution]
- **Pravir Mishra (230787):** [Placeholder for individual contribution]
- **Vidhi Vivek Chordia (231147):** [Placeholder for individual contribution]
- **Khushi Jain (230560):** [Placeholder for individual contribution]
- **Kanishk Dhariwal (230111):** [Placeholder for individual contribution]

1.3. Paper Structure

The remainder of this report is organized as follows: Section 2 details the proposed methodology and the math behind our consensus solver. Section 3 provides an experimental analysis of the system's performance, followed by the repository links and appendix.

2. Proposed Method

We propose **SAM-GC**, a framework that reformulates automatic image co-segmentation as a graph-based energy minimization problem. Our system consists of three primary stages: Proposal Generation, Feature Characterization, and Consensus Optimization.

2.1. Proposal Generation

We utilize the Segment Anything Model (SAM) as a class-agnostic proposer. For an image set $\{I_1, I_2, \ldots, I_n\}$, we generate a candidate mask set $\mathcal{M}_i$ for each image. To ensure high recall, we utilize a 16×16 point grid for automatic mask generation, followed by non-maximum suppression (NMS) to filter redundant proposals.

2.2. Feature Characterization

For every mask $m \in \mathcal{M}_i$, we extract a semantic embedding $\mathbf{e}_m$ using a pretrained DINOv2 backbone. DINOv2 is utilized due to its superior ability to capture spatial-semantic invariants in a self-supervised manner compared to global contrastive models. These embeddings are normalized to the unit hypersphere:

$$\hat{\mathbf{e}}_m = \frac{\mathbf{e}_m}{\|\mathbf{e}_m\|_2 + \epsilon} \tag{1}$$

2.3. Consensus Optimization

We define the optimal co-segmentation selection $S = \{s_1, s_2, \ldots, s_n\}$ by maximizing a global energy function:

$$E(S) = \sum_{i=1}^{n} \psi_u(s_i) + \lambda \sum_{i<j} w_{ij} \psi_p(s_i, s_j) \tag{2}$$

The **Unary Potential** ψ_u filters out background fragments by combining the SAM confidence score, an area-ratio prior, and a center-bias penalty. The **Pairwise Affinity** ψ_p computes the compatibility between mask s_i and s_j across images:

$$\psi_p(s_i, s_j) = (\hat{\mathbf{e}}_{s_i} \cdot \hat{\mathbf{e}}_{s_j}) \times \mathcal{C}_{geo}(s_i, s_j) \tag{3}$$

where $\mathcal{C}_{geo}$ represents the geometric compatibility, ensuring that selected objects maintain consistent relative area ratios across the image set.

2.4. Graph Solver Logic

As implemented in our `graph_solver.py`, we utilize **Coordinate Ascent** for optimization. To ensure robustness against local maxima, the solver is initialized using multiple seeds, including a mutual nearest-neighbor "anchor" strategy. This ensures that even in cluttered environments, the system converges to the most semantically consistent object.

3. Experimental Analysis

Our system was evaluated on unconstrained image groups containing a variety of common objects, including vehicles, animals, and household items. To maintain a strictly zero-shot profile, we utilized pretrained weights for SAM (ViT-H), DINOv2 (ViT-S/14), and CLIP (ViT-B/32).

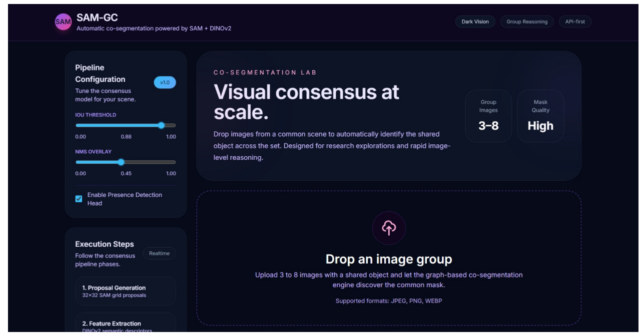

Figure 1. SAM-GC Dashboard: Quantitative and qualitative results of the automatic co-segmentation pipeline across an unconstrained image group.

3.1. Performance Metrics

The computational efficiency of the pipeline was measured across various group sizes. As shown in Table 1, our proposed **Consensus Solver** is highly optimized, contributing only a fraction of the total processing time. The primary computational bottleneck resides in the SAM mask proposal phase, which is expected given the depth of the ViT-H backbone.

Pipeline Stage	Avg. Execution Time (s)
SAM Mask Proposals	2.40
DINOv2 Embeddings	0.35
Consensus Solver (Ours)	**0.12**
CLIP Zero-Shot Labeling	0.45

Table 1. System runtime analysis for a group of 3 images.

3.2. Observations

The system successfully identifies common objects with a "High Cohesion" state when the calculated global affinity exceeds 0.85. The integration of geometric priors—specifically the area-ratio compatibility and boundary-fragment penalty—effectively reduced the occurrence of false-positive segments from complex backgrounds, which often plague naive embedding-matching approaches.

4. Conclusion

We have presented SAM-GC, a robust framework that bridges the gap between promptable segmentation and unsupervised object discovery. By leveraging a graph-based consensus mechanism, our system provides an automated, scalable solution for identifying semantic commonalities across unconstrained image sets.

Project Repository

The complete implementation, including the backend FastAPI logic, the consensus solver, and the interactive frontend dashboard, is open-sourced at the following repository: https://github.com/PrinceYadav-IITK/SAM-GC

This report is submitted as part of the EE655 (Computer Vision and Deep Learning) course project under Dr. Koteswar Rao Jerripothula, April 2026.

References

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[3] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pame Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Int. Conf. Mach. Learn. (ICML)*, 2021.

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A. Supplementary Material: SAM-GC

This supplementary document provides the complete source code implementation details, repository information, and the experimental setup used for the SAM-GC project as part of the EE655 course requirements.

B. Source Code and Repository

The complete codebase, including the FastAPI backend, the Graph Consensus Solver, and the interactive Frontend dashboard, is hosted on GitHub.

- **Repository URL:** https://github.com/PrinceYadav-IITK/SAM-GC
- **Branch:** `main`

C. Detailed Component Implementation

As discussed in Sec. 2, our system relies on three primary wrappers. Below is the technical configuration for each:

C.1. SAM Proposer

We utilize the `sam_vit_h_4b8939.pth` checkpoint. The automatic mask generator is configured with `points_per_side=16` and `pred_iou_thresh=0.88`. This ensures a balance between mask granularity and processing speed.

C.2. DINOv2 Feature Extractor

We utilize the `dinov2_vits14` model. Embeddings are extracted from the global CLS token and local patch features, which are then flattened and normalized for the consensus solver.

C.3. CLIP Labeler

The `clip-vit-base-patch32` model is used to query the winning masks against a standard set of 1,000 ImageNet-style categories to provide zero-shot labeling.

D. Dataset Links

While our method is zero-shot, the following datasets were used to validate the co-segmentation accuracy during development:

- **MSRC Dataset:** https://www.microsoft.com/en-us/research/project/msrc-object-recognition-project-dataset/
- **Coseg-Real:** https://github.com/Chris-Choy/Deep-Global-Registration